Light-Dependent Reactions

Unit 3C Slides 14 through 19: Honors High School Biology

Slide 14: Light-Dependent Reactions

We are now focusing on the first major stage of photosynthesis. The light-dependent reactions capture energy from photons and convert it into ATP and NADPH. These products do not leave the chloroplast as food. They provide short-term chemical energy and high-energy electrons for the Calvin cycle.

The reactions occur in the thylakoid membrane, where pigments, electron carriers, and ATP synthase are arranged to transfer energy in a specific sequence. Water supplies replacement electrons, and oxygen is released as a byproduct.

Slide 15: Light Reactions Overview

The light reactions begin when chlorophyll and accessory pigments absorb photons. This energy excites chlorophyll electrons, allowing them to enter a series of membrane proteins. This energy ultimately produces ATP and NADPH.

Two photosystems participate. A photosystem is a large complex of proteins and pigment molecules arranged around a reaction-center chlorophyll. Photosystem II functions first, despite its number, and Photosystem I functions later. The numbering reflects the order in which scientists discovered them rather than their order in the pathway.

Slide 16: Light Reactions and Electron Transport

The light reactions resemble cellular respiration because both use electron transport and a proton gradient to drive ATP synthase. The source of electron energy differs. Cellular respiration begins with chemical energy from food, while photosynthesis uses light to raise electrons to higher energy levels.

The slide labels two carrier sequences as ETC 1 and ETC 2. In the first sequence, Photosystem II supplies excited electrons to carriers that help build an H+ gradient and support ATP production. Photosystem I then absorbs another photon, re-energizes the electrons, and transfers them through additional carriers to NADP+ reductase. This enzyme reduces NADP+ to NADPH.

ATP and NADPH then move into the stroma, where the Calvin cycle uses them. ATP provides energy, while NADPH provides high-energy electrons and hydrogen needed to reduce carbon compounds.

Slide 17: Light Reactions Video

As you watch, track three related movements. Follow the electrons from water through Photosystem II and Photosystem I. Follow hydrogen ions as they accumulate inside the thylakoid. Finally, identify where ATP and NADPH form on the stroma side of the membrane.

Do not try to memorize every carrier protein from the animation. Concentrate on the larger energy pattern: light excites electrons, electron transfer helps build a proton gradient, and the gradient powers ATP synthase.

Slide 18: Step 1 - Photosystem II and ETC 1

The pathway begins when pigments in Photosystem II absorb light and transfer the energy to the reaction-center chlorophyll, P680. Two excited electrons leave chlorophyll and enter the electron transport chain. Because chlorophyll has lost electrons, the photosystem needs a replacement source.

An oxygen-evolving complex splits water. This process supplies electrons to Photosystem II, releases H+ into the thylakoid interior, and produces oxygen gas. The oxygen atoms in released O2 therefore come from water, not carbon dioxide.

As electrons move through the first transport chain, their energy helps pump additional H+ from the stroma into the thylakoid interior (the lumen). This produces a high concentration of H+ inside the thylakoid. H+ then flows down its electrochemical gradient through ATP synthase and back into the stroma. ATP synthase uses that flow to phosphorylate ADP, producing ATP.

Slide 19: Photosystem II and ETC 1 Diagram

Use the diagram to identify the membrane orientation before following the arrows. The thylakoid interior is inside the membrane-bound compartment, while the stroma surrounds the thylakoid. Photosystem II and the electron carriers span the thylakoid membrane.

On the left, water splits near Photosystem II. Electrons replace those lost by chlorophyll, oxygen forms, and H+ enters the thylakoid interior. Across the membrane, electron transfer helps increase the H+ concentration inside the thylakoid. In the ATP synthase section, H+ returns to the stroma and ATP forms from ADP and phosphate.

Notice that the electron pathway and proton pathway are connected but different. Electrons travel through membrane carriers. Hydrogen ions move across the membrane and then through ATP synthase. The gradient, not direct contact with electrons, powers ATP synthase.

Closing Statement

Photosystem II uses light and water to begin electron flow. The first transport chain captures part of the electrons' energy in a proton gradient, and chemiosmosis converts that stored energy into ATP. The electrons then reach Photosystem I, where light re-energizes them to produce NADPH.